Back to the Future: Opening The Hood Of Everything

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Abstract

The early Internet empowered users to shape their online experiences, from editing websites in Netscape Communicator to customizing MySpace profiles. Today, major platforms like Facebook and TikTok limit user control, reducing participation to content creation rather than system customization. This shift contrasts sharply with Alan Kay's 1984 vision of user-editable tools and Douglas Engelbart's goal of augmenting human intellect. This position paper argues that recent advances in generative AI, particularly large language models, offer a path to reclaiming user agency. These models enable code generation, customization, and interaction paradigms that could allow users to "open the hood" of software, aligning with participatory design principles. By combining generative AI with deep user involvement, we can reimagine human-computer interaction in a way that enhances autonomy and creativity. This paper advocates for a future where users can edit, shape, and redefine their digital tools.

CCS Concepts

• **Human-centered computing → User interface programming**.

Keywords

User Agency, Participatory Design, Generative AI, End-user programming, Customizable software.

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1 Motivation

The Internet was malleable and fun when I started using computers in 1998. Using Netscape Communicator, I could edit any website by clicking just one button. This allowed me to understand how websites worked. That is how I learned HTML, CSS, and Javascript. When I got a little bit older, early social media platforms like MySpace emerged. Users could edit their profile pages freely on MySpaces, adding HTML, graphics, and music to express themselves.

Today, users grow up with apps and platforms like Facebook and TikTok. On Facebook, they have minimal control over their profile pages. On TikTok, they can express themselves using video content but cannot change the platform or the recommendation algorithm. How Facebook and TikTok as platforms work cannot be influenced by users. The same applies to smartphones, regardless of whether people use Android or iOS.

At the risk of sounding like an old man: In hindsight, the Internet that I used as a child felt like growing up near a forest, where I could use the wood of the forest to build my own toys to entertain myself. Children today grow up with technology that feels like a sterile mall. They can visit Amazon's shops. They can also participate in the occasional talent show called TikTok.

2 Back to the Future

The limits of people's control over contemporary software like smartphone apps and social media platforms are especially obvious when we consider the early vision of computing's pioneers. In 1984, Alan Kay wrote an essay titled "Opening the Hood Of A Word Processor" [11]. Kay argued:

> As we become more sophisticated, our ideas about what our word processors, our graphics systems should do for us diverge more and more from those of the initial designers. We now want to edit our tools as we have previously edited our documents.

Kay envisioned letting users "open the hood" and customize software in ways that applications programmers didn't specifically anticipate. He also introduced the important concept of a "power to simplicity ratio". He describes how people can derive their own spreadsheet- and word-processing tools from first principles. Forty years later, the vision still remains precisely this: a mere vision.

In this position paper, I argue why this is a lost opportunity, why such past visions were not radical and user-friendly enough, and how this can be fixed. I believe that advances in artificial intelligence, namely large language models, could be the missing link that could enable users to open the hood of everything, not just their word processors and spreadsheets. I envision extending this paradigm to websites, spam filters, and even social media platforms.

3 History

In his 2022 book "Human-Centered AI", Ben Shneiderman argued:

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> "Human-Centered Al (HCAI) enables people to see, think, create, and act in extraordinary ways, by combining potent user experiences with embedded AI support services that users want."

This mirrors the original vision of Douglas Engelbart, put forward in his visionary text "Augmenting Human Intellect: A Conceptual Framework" from 1962. In it, Engelbart [4] described augmenting the human intellect as "increasing the capability of a man to approach a complex problem situation, to gain comprehension to suit his particular needs, and to derive solutions to problems."

I think that back in the day, the missing link were insufficient capabilities of artificial intelligence techniques and a lack of participatory design. Kay's vision was driven by mathematical thinking. While this is obviously and demonstrably powerful, based on my lived experience, I argued that this is too complex and abstract for most people.

As human-computer interaction scholars, I believe that we should imagine ways to make it easy to open the hood of all apps and websites. Editing software needs to be easy and obvious, as it was for me to edit my first websites using Netscape Communicator in 1998.

4 Generative Artificial Intelligence

Artificial intelligence has a long history. Definitions range from Tesler's Theorem as "AI is whatever hasn't been done yet" [7] to Goodfellow et al.'s definition that describes artificial intelligence (A.I.) as a field with many practical applications and active research topics including, but not limited to, intelligent software to automate routine labor, understand speech or images, make diagnoses in medicine and support basic scientific research [5]. Machine learning, especially artificial neural networks and so-called deep learning, has led to recent advances in machine translation, object recognition, speech recognition, autonomous driving, medical diagnostics, and game playing [6, 10]. However, the steady progress should not overshadow the paradigm shift enabled by recent advances in foundation and large language models.

With the advent of machine learning, how tasks are executed is inferred from examples. Deep learning extends this by enabling the automatic extraction of high-level features for such predictions. Foundation models take this further, allowing complex capabilities like in-context learning to emerge. Bommasani et al. define the term foundation models as "any model that is trained on broad data (generally using self-supervision at scale) that can be adapted (e.g., fine-tuned) to a wide range of downstream tasks" [2].

An exciting recent advance in large language models are so-called emergent properties. Emergent properties refer to capabilities that arise unpredictably as model scale increases, often appearing non-linearly [16]. Notable examples include in-context learning [3], where models generalize from a few prompt examples without explicit retraining, and zero-shot/few-shot learning, enabling such large language models to perform tasks without prior task-specific training. Additionally, multi-step reasoning and chain-of-thought [17] prompting improve their problem-solving abilities, while theory of mind-like reasoning allows models to infer beliefs and perspectives. Notable examples of emergent capabilities include translation between unseen languages and analogical/metaphorical thinking, demonstrating higher-level abstraction abilities. Some LLMs also exhibit rudimentary strategic planning, predicting multi-step outcomes in text-based tasks. These properties highlight the complex, non-linear development of intelligence-like behaviors in LLMs and underscore the need for further investigation into their origins, limitations, and potential applications.

An especially noteworthy and currently underexplored application of generative AI is their ability for code generation and debugging [15], where models produce and refine executable programs. These advances will lead to a significant paradigm shift in designing human-computer interaction. Prior already documented first steps in this direction. Vaithilingam et al., for instance, shows how natural language and dynamically synthesized UI widgets can be adapted to support users who are editing visualizations [14]. Masson et al. follow a similar approach by combining generative artificial intelligence with the principles of direct manipulation [12]. This includes a continuous representation of objects of interest, reusing prompt syntax, manipulable outputs to compose or control the effect of prompts, and undo mechanisms.

These advances put us in a unique position to understand and augment human thinking. I believe these advances can enable us to realize Kay's original vision. They can allow users to open the hood of everything, especially if we, as HCI scholars, act as primus inter pares in participatory design settings, guiding the design and development but treating everyone's input as equally valuable.

5 Participatory Design

My position is participatory design [1], and deep user involvement in line with the so-called co-creation [8, 9] can help overcome the challenges associated with designing novel human-AI interactions and thus augment human thinking by protecting human autonomy. I think the deep involvement and mutual learning between researchers and users can also help us understand and assess the impact of generative AI systems. Thus, it can be ensured that human skills and workflows are augmented, not replaced.

I position myself in the Scandinavian participatory design tradition as described by Bødker et al. [1]. In this tradition, participatory design is committed to democracy at the workplace and beyond and is concerned with the empowerment of people through the processes of design. They view participatory design as an emancipatory practice rooted in mutual learning between designers and people. Bødker et al. argue that this is done by seeing human beings as skillful and resourceful in the development of their future practices. My position is informed by my lived experience trying to facilitate such deep user involvement.

The deep involvement of humans is central. Following Winograd and Flores [18], "We encounter the deep questions of design when we recognize that in designing tools we are designing ways of being." Therefore, the guiding principle for designing GenAI systems has to be deep user involvement since users are the experts in who they are and who they want to be.

For HCI and AI, this means we need to go back to the future. Surprisingly, this deep user involvement in shaping their tools has a successful but somewhat forgotten history. However, as an old man, I remember how central features of platforms like Twitter were invented by users [13]. On August 23rd, 2007, Chris Messina tweeted: "how do you feel about using # (pound) for groups. As in

#barcamp [msg]?". He invented what is known as the hashtag. The practice of replying to a user by using the @-symbol was first done by Robert Andresen, who tweeted "@ buzz - you broke your thumb and youre still twittering? that's some serious devotion". Eric Rice started the practice of retweets by directly copying a message from somebody. These examples illustrate the power of human thought and the potential of tools for thought. They demonstrate the vast potential of deeply involving users in the design of their software and their platforms. I think combining the power of GenAI and the power of participatory design has the potential to allow users to edit their tools as they have previously edited their documents, fulfilling Alan Kay's vision.

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